

Zofia Chmiel

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EDUCATION

Duke University

Durham, NC

B.S. in Mechanical Engineering with Marine Science & Conservation Minor (GPA: 3.3/4.0)

Aug. 2023 – May 2027

Relevant Coursework: Dynamics • Thermodynamics • Mechanics of Materials • Linear Algebra • Mechatronics • Controls • Heat & Mass Transfer

Activities: Pratt Tour Guide, Catholic Center Liturgy Director, Theta Tau, Kappa Alpha Theta, Society of Women Engineers, Duke Club Swimming

SKILLS & INTERESTS

Software Skills: SolidWorks, Autodesk Fusion, Onshape, AutoCAD, Python, HTML, MATLAB, Bluebeam, Microsoft Office

Engineering Skills: CNC Machining, 3D Printing, Laser Cutting, Rapid Prototyping, Woodworking, Welding

Interests: Scuba Diving, Running, Historical Fiction & Mystery Novels, Collaging, Interior Design, Thrifting

FIELD ENGINEERING

Hardware Engineering Intern | FaunaLabs - Durham, NC

January 2026 - present

- Contributed to the mechanical redesign of FaunaTag, a biometric wearable for whale health monitoring, using CAD to improve shape, durability, waterproofing, and deployment reliability in harsh marine environments for long-term, field-ready physiological data collection
- Collaborated with a multidisciplinary team to design, build, and test ~5 prototype iterations (approx. 6" long, 1.5 lbs. each) integrating IMU, hydrophone, and bio-optical (fNIRS-style) sensors for physiological and behavioral tracking.
- Refined the suction-cup attachment and sensor housing to maintain secure, non-invasive skin contact while minimizing disruption to whale behavior.

Mechanical Design Lead | Plastic Bioremediation Research - Durham, NC

January 2026 - present

- Designed and prototyped a bioreactor system that grinds household plastic waste to a uniform size for microbial breakdown.
- Mutated plastic-degrading bacterial strains in lab settings to identify variants that more effectively decompose plastic and consume carbon content.
- Integrated mechanical and biological subsystems, including cell culture handling and fluid control design, to build a scalable system for converting household plastic waste into compostable material.

Global Facilities & Engineering Intern | Boehringer Ingelheim - Ridgefield, CT

May 2026 – August 2026

- Investigated vacuum pump reliability issues and led vendor walkthroughs for a ~\$300K KalWall blast panel replacement project, plus heat exchanger and water softener design planning to improve reliability and freeze protection in a GMP pharmaceutical facility.
- Performed HVAC air exchange and pressure calculations across 45 process rooms, identified performance gaps and coordinating with facilities and lab teams to support critical punch-list closeout ahead of GMP cleaning and turnover.
- Built HTML prototypes for an AED locator tool and an equipment disposal workflow, supported a published site rollout.

Estimator Intern | Saunders Construction - Denver, CO

May 2025 - August 2025

- Led preconstruction for an ~\$8M structural facility, managed cost estimation, trade coordination, and risk mitigation across civil, structural, and architectural scopes to ensure alignment with budget targets, technical requirements, timelines, and feasibility from concept to pricing.
- Created detailed bid tabs, cost models, and scope matrices using Excel and Bluebeam to support pricing accuracy, subcontractor alignment, and constructability review, contributing to risk mitigation, competitive bidding, and effective decision-making during the preconstruction process.
- Presented value engineering recommendations and collaborated with engineers to align with technical, budget, and schedule requirements.

LEADERSHIP

Project Manager | Durham, NC & Eswatini

August 2024 - September 2025

- Managed the design and planning of a suspended pedestrian bridge and WASH system serving 300+ residents, overseeing stakeholder coordination and ensuring alignment with engineering standards, environmental constraints, and long-term functionality goals.
- Partnered remotely with engineers, NGOs, and local officials to develop structural plans, perform site feasibility assessments, and ensure safe, sustainable delivery of bridge and WASH infrastructure aligned with technical requirements and community needs.

In Country Project Manager | Kaihura, Uganda

September 2023 - July 2024

- Led an 11-student team to complete an 8-week rural school construction project; managed daily operations, coordinated material logistics, and communicated effectively with local laborers to ensure alignment with design specifications, schedule goals, and safety protocols.
- Adapted construction plans to overcome material shortages and delays while ensuring alignment with project specs and community needs.